



FINE-TUNED UNDERSTANDING: ENHANCING SOCIAL BOT DETECTION WITH TRANSFORMER-BASED CLASSIFICATION

DEPARTMENT OF INFORMATION
TECHNOLOGY

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INTRODUCTION

- Malicious social bots impersonate humans, spread misinformation, and threaten digital discourse integrity on platforms like Twitter.
- Differentiating bot-generated text from human-written content is increasingly difficult due to advanced neural language models .
- We propose a transformer-based approach that fine-tunes Pre-trained Language Models on bot-specific tweet data, combined with a Feedforward Neural Network for binary classification.

PROBLEM IDENTIFICATION

- **Increasing bot sophistication** – Modern social bots closely mimic human language, making detection difficult.
- **Low accuracy of traditional methods** – Rule-based and feature-engineered models fail against advanced bots.
- **Poor contextual understanding** – Conventional approaches cannot capture deep linguistic and semantic patterns.
- **Manual feature engineering** – Designing features is time-consuming, domain dependent, and error-prone.
- **Limited adaptability** – Models struggle to generalize across different datasets and evolving bot behaviors.

ACHIEVEMENT OF OBJECTIVES

- **Transformer-based Fine-Tuning** Employed BERT, RoBERTa, DistilBERT, XLM-RoBERTa fine-tuned on bot-specific datasets (TweepFake & fox8-23) for contextual tweet embeddings
- **Tweet-Level Bot Classification** Designed a Feedforward Neural Network (FNN) classifier achieving binary classification (Human vs. Bot) with high precision and recall
- **Explainability Integration** Integrated SHAP, BERT Visualizer, and LLM-based prompting to interpret model decisions and enhance transparency

ACHIEVEMENT OF OBJECTIVES

- **Comparative Benchmarking** Systematically evaluated 6 PLMs against traditional embeddings (Word2Vec, GloVe + BiLSTM) across two diverse datasets
- **Domain Adaptation** Addressed the gap of bot-specific pre-training by fine-tuning on specialized bot-generated content datasets

DATA COLLECTION & PREPROCESSING MODULE

- Collects social media data containing human and bot-generated content
- Removes noise such as URLs, hashtags, emojis, and special characters
- Performs tokenization and text normalization
- Prepares clean and structured input for model training

TRANSFORMER-BASED FEATURE EXTRACTION MODULE

- Uses pre-trained transformer models like BERT or RoBERTa
- Generates contextual and semantic text embeddings
- Captures linguistic patterns used by advanced social bots
- Converts text into meaningful numerical representations

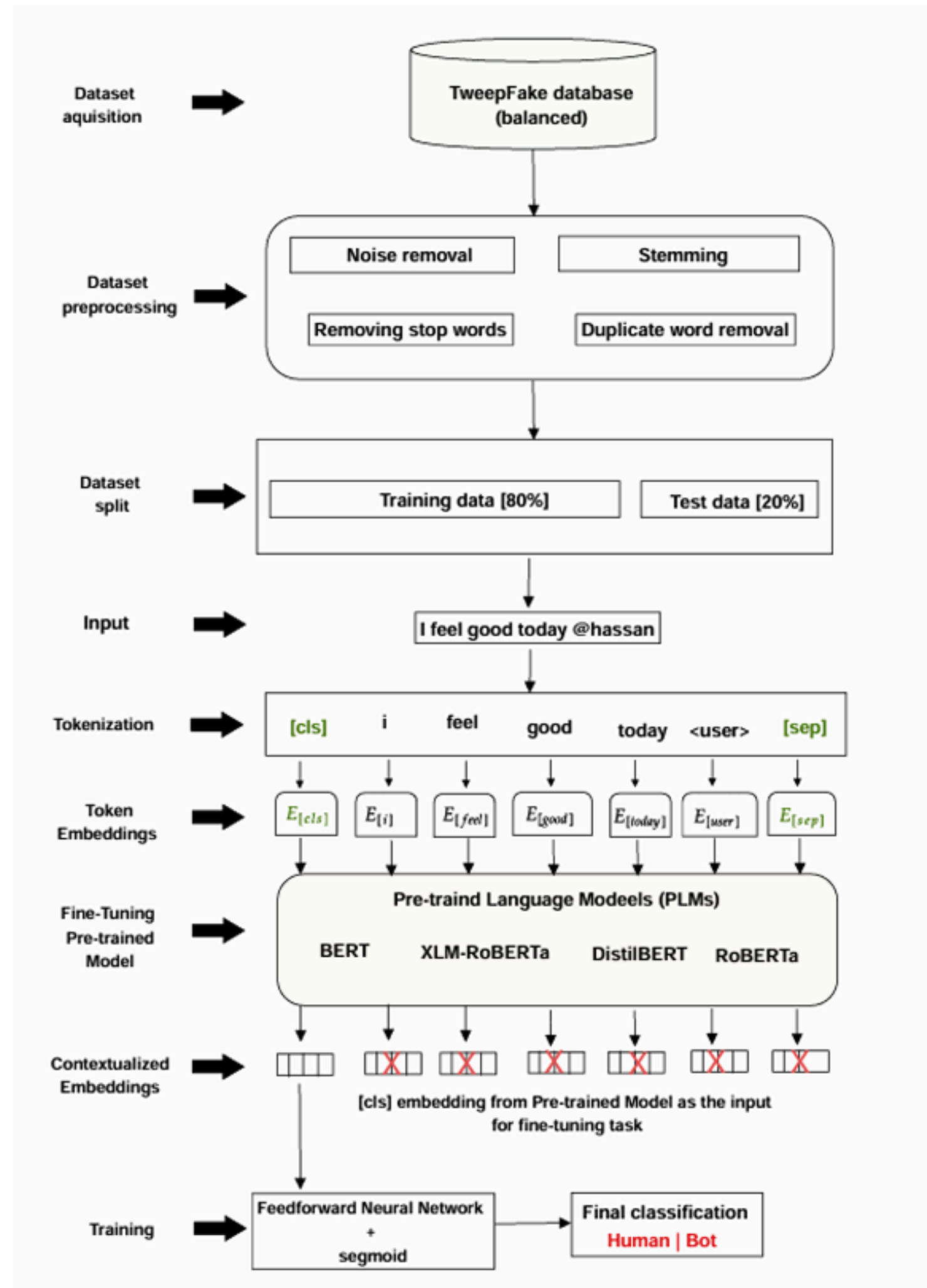
FINE-TUNING & CLASSIFICATION MODULE

- Fine tunes pre-trained language models on bot detection datasets
- Trains a deep neural network for binary classification
- Classifies content as human-generated or bot-generated
- Improves detection accuracy and generalization

EVALUATION & RESULT ANALYSIS MODULE

- Evaluates model performance using accuracy, precision, recall, and F1-score
- Analyzes confusion matrix and classification results
- Compares results with traditional detection methods
- Validates system effectiveness and reliability

ARCHITECTURAL DIAGRAM



DATASET COMPARISON

Dataset	Total Tweets / Accounts	Human Tweets	Bot Tweets	Description
TweepFake	25,572 tweets	12,786	12,786	Balanced dataset containing tweets generated by both humans and bots. Bot tweets are generated using models such as Markov Chains, RNN, LSTM, and GPT-2.
fox8-23	368,028	218,245	149,783	Large-scale Twitter dataset containing real human accounts and bot accounts used for benchmarking bot detection systems.



LITERATURE SURVEY

Paper Title	Journal / Venue	Algorithm(s)	Environmental Setup	Advantages	Disadvantages
Detecting Social Bots on Twitter Using Deep Learning (2019)	IEEE Access	CNN, LSTM with GloVe embeddings	Tweet-level classification on Twitter datasets	Captures temporal and textual patterns	Limited contextual understanding
Bot Detection Based on Content and Metadata (2020)	WWW Conference	LSTM + metadata fusion	User- and tweet-level Twitter data	Combines behavioral and textual features	Poor generalization to unseen bots
Transformer-Based Social Bot Detection (2021)	ACM CIKM	BERT fine-tuning	Cresci-2017 Twitter bot dataset	Strong contextual representations	High computational cost
Fine-Tuned Pretrained Language Models for Bot Detection (2023)	IEEE Access	BERT, RoBERTa, DistilBERT	Tweet-level binary classification	High F1-score without feature engineering	Limited model interpretability
Explainable Transformer Models for Social Bot Detection (2024)	arXiv preprint	BERT + SHAP explainability	Twitter bot datasets with attention analysis	Improved transparency and trust	Additional computational overhead

Paper Title	Journal / Venue	Algorithm(s)	Environmental Setup	Advantages	Disadvantages
Multimodal Social Bot Detection Using Transformers (2023)	Elsevier Knowledge-Based Systems	BERT + visual feature fusion	Twitter dataset with text and profile metadata	Captures richer bot behavior patterns	Requires multimodal data availability
Generalization-Aware Transformer Models for Bot Detection (2023)	EMNLP	RoBERTa with domain adaptation	Cross-dataset Twitter bot evaluation	Better robustness to unseen bot types	Performance drops with noisy labels
Detecting AI-Generated Social Media Text Using LLMs (2024)	AAAI Conference	GPT-based detectors, RoBERTa	Human vs LLM-generated tweet datasets	Effective against neural text generators	Struggles with mixed human-bot content
Explainable Social Bot Detection via Attention Analysis (2024)	IEEE Transactions on AI	Transformer attention visualization	Tweet-level classification with explainability layer	Improves interpretability of predictions	Adds inference complexity
Fine-Tuned Transformer Models for Cross-Platform Bot Detection (2025)	arXiv preprint	BERT, XLM-RoBERTa fine-tuning	Twitter and Weibo datasets	Supports multilingual and cross-platform detection	High computational and training cost

DEMONSTRABLE RESULTS

Tue Mar 10 14:18

Explainable AI Bot Detection

localhost:8501

Deploy

Explainable AI Bot Detection Dashboard

Enter Tweet

The post COVID-19 era will bring with it several opportunities and India's track record convinces me we will rise to the occasion and embrace the opportunities ahead. #AatmanirbharBharat
<https://t.co/ymqe3bjiml>

Run Explainable AI

AI Processing Pipeline

User Tweet
↓
Preprocessing
↓
Tokenization
↓
Token IDs
↓
Transformer Encoder
↓
Hidden States
↓
Neural Network

DEMONSTRABLE RESULTS

Explainable AI Bot Detection

localhost:8501

Deploy

Preprocessing

Original

The post COVID-19 era will bring with it several opportunities and India's track record convinces me we will rise to the occasion and embrace the opportuniti

URL Removed

The post COVID-19 era will bring with it several opportunities and India's track record convinces me we will rise to the occasion and embrace the opportuniti

HTML Removed

The post COVID-19 era will bring with it several opportunities and India's track record convinces me we will rise to the occasion and embrace the opportuniti

Special Characters Removed

The post COVID era will bring with it several opportunities and Indias track record convinces me we will rise to the occasion and embrace the opportunities a

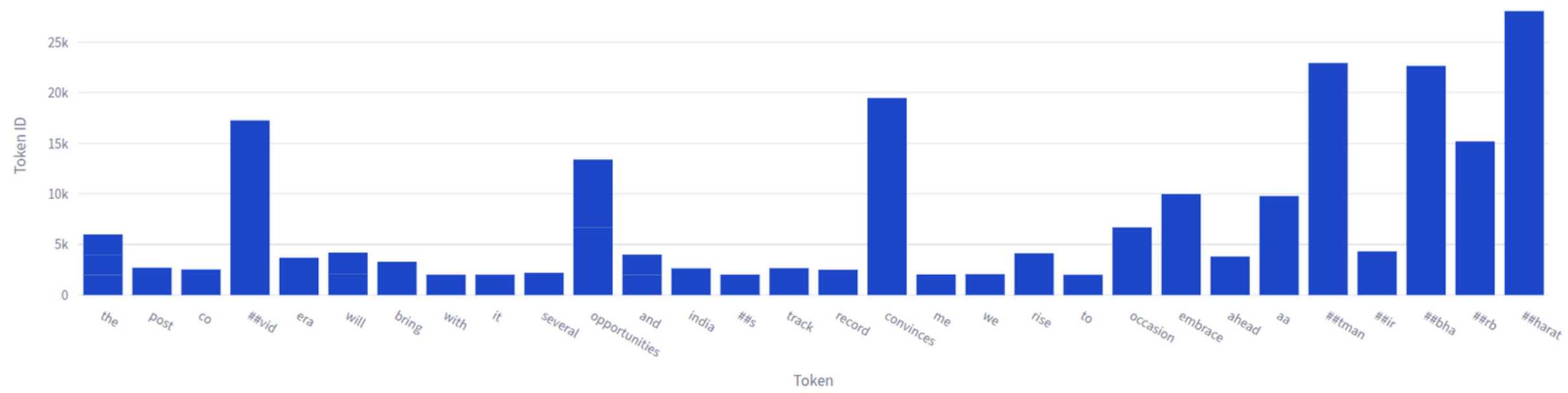
Lowercase

DEMONSTRABLE RESULTS

abc Tokenization

	Token	Token ID
0	the	1996
1	post	2695
2	co	2522
3	##vid	17258
4	era	3690
5	will	2097
6	bring	3288
7	with	2007
8	it	2009
9	several	2195

Token ID Distribution



DEMONSTRABLE RESULTS

Attention Flow Across Layers

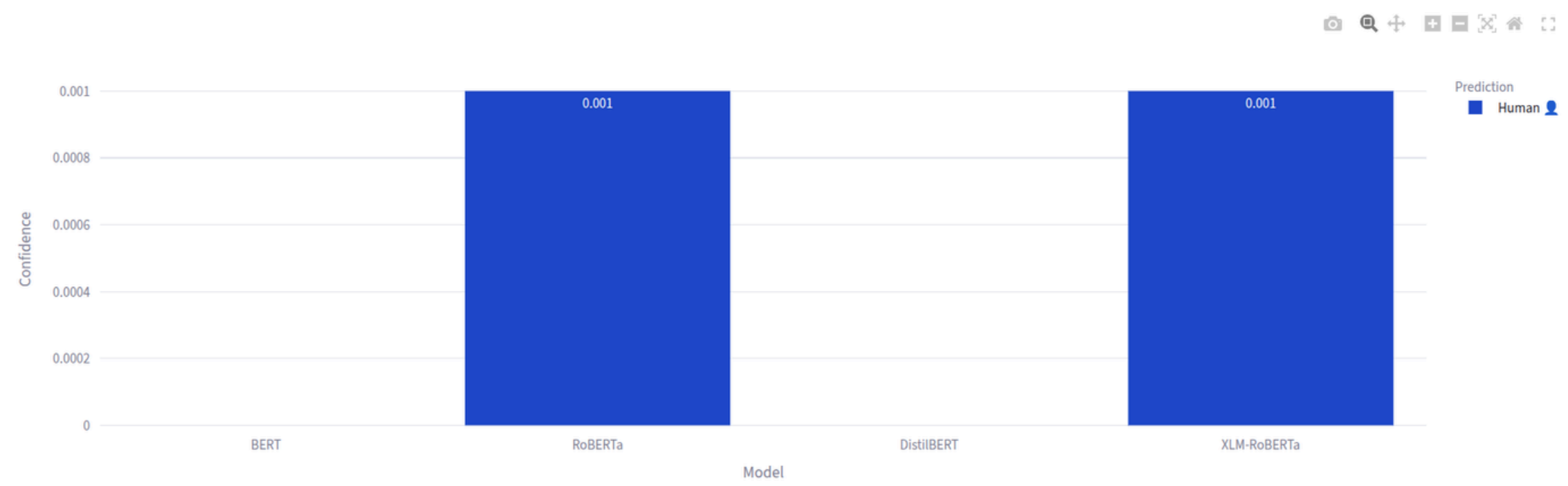
BERT Attention Flow



DEMONSTRABLE RESULTS

Model Comparison

	Model	Prediction	Confidence
0	BERT	Human 	0
1	RoBERTa	Human 	0.001
2	DistilBERT	Human 	0
3	XLM-RoBERTa	Human 	0.001



Final Prediction

HUMAN ACCOUNT DETECTED 

DEMO VIDEO LINK

https://drive.google.com/file/d/1NnMIFgrRHZSJZpP_Tl7d80AWY3QY_ZO3/view?usp=sharing

SOCIETAL IMPACT

- **Combating Misinformation** Reduces the spread of fake news, propaganda, and manipulated content by identifying bot-generated tweets at scale
- **Economic Integrity** Reduces fraud, fake engagement, and manipulative marketing - creating fairer digital economies for creators and businesses
- **Protecting Vulnerable Users** Shields younger audiences, elderly users, and marginalized communities from targeted bot campaigns, scams, and harmful content
- **Authentic Human Connection** Enables users to engage with real people, fostering genuine relationships and meaningful dialogue

REQUIREMENTS

Hardware

- Minimum: Intel i5 / Ryzen 5, 8 GB RAM, 100 GB SSD
- Recommended: 16 GB RAM, 4 GB GPU, SSD storage

Software

- Models: BERT, RoBERTa, DistilBERT, XLM-RoBERTa, GPT-based embeddings
- Development: Python, PyTorch, Hugging Face

Datasets

- Social Bot Data: Twitter bot datasets (e.g., fox8-23, TweepFake)
- Synthetic Data: AI-generated bot tweets
- Labels: Human / Bot (tweet-level)

Community Support

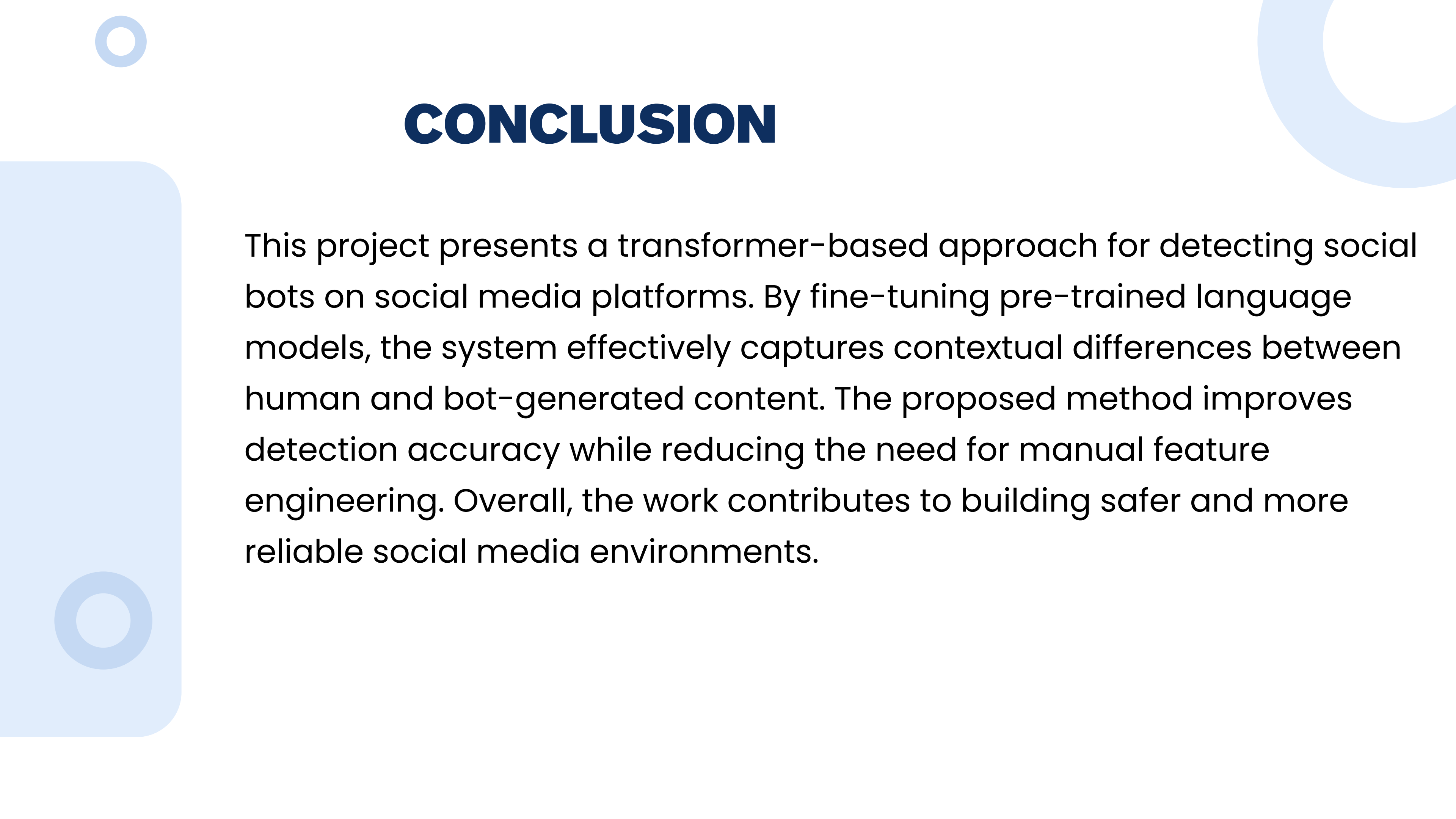
- Tools: Hugging Face Hub, GitHub
- Environment: Local GPU / Google Colab
- Explainability: SHAP, attention visualization

FUTURE ENHANCEMENTS

- Multilingual PLMs for cross-language bot detection
- Combine content features with social graph data using TwiBot-20/22 datasets.
- Optimize models for real-time, low-latency deployment on edge/cloud platforms.
- Build adaptive online learning models to counter evolving AI-generated bots.

CHALLENGES

- **Computational Cost:** Use DistilBERT to reduce GPU usage.
- **Short Text Limit:** Optimize model for short tweets (≤ 280 chars).
- **Bot Evolution:** Use periodic retraining to detect new bots.
- **Domain Shift:** Support multiple platforms and languages.
- **Privacy & Ethics:** Ensure transparent and privacy-safe detection.



CONCLUSION

This project presents a transformer-based approach for detecting social bots on social media platforms. By fine-tuning pre-trained language models, the system effectively captures contextual differences between human and bot-generated content. The proposed method improves detection accuracy while reducing the need for manual feature engineering. Overall, the work contributes to building safer and more reliable social media environments.

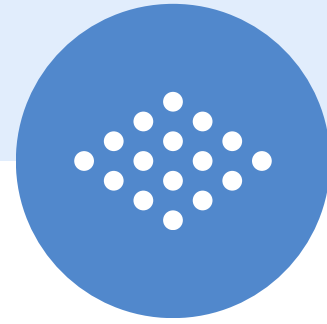


Q/A

REFERENCE:

Fine-Tuned Understanding: Enhancing Social Bot Detection with Transformer-based Classification -

<https://ieeexplore.ieee.org/document/10630818>



**THANK
YOU!**

